

INSTITUTIONAL FINANCIAL INFRASTRUCTURE

Corporate Profile & Platform Capabilities

A comprehensive guide to M&F Technologies' high-performance core lending systems, automated risk decisioning engines, and bank-grade transactional middleware engineered for commercial banks, SACCOs, and digital microfinance institutions.

\$1.4B+

PROCESSED VOLUME

99.99%

PRODUCTION UPTIME

<100ms

P99 API LATENCY

Headquarters: The Pavilion, 4th Fl, Lower Kabete Rd, Nairobi, Kenya**Operations:** Pan-African Deployments across 12+ Jurisdictions**Compliance:** SOC 2 Type II Certified • ODPC & GDPR Compliant**Corporate Contact:** info@mfttechnologies.org**Institutional Hotline:** +254 748 329 410**Official Web Portal:** <https://mfttechnologies.org>



• FOUNDATIONAL PEDIGREE & MISSION

Powering the Future of Institutional Credit

M&F Technologies is an institutional financial technology company that engineers, deploys, and operates mission-critical lending infrastructure for commercial banks, credit unions, and microfinance providers across emerging markets.

Company Overview & Market Position

Founded in 2021, M&F Technologies addresses the acute structural gap between rigid, legacy core banking architectures and the soaring demand for instantaneous, automated digital credit. By pairing mathematical double-entry ledger integrity with event-driven microservices, we enable licensed financial institutions to originate, underwrite, disburse, and collect loans with zero balance drift and sub-second response times.

Our infrastructure serves as the transactional backbone for commercial bank retail portfolios, community SACCOs, and high-volume microfinance operators, processing billions in transactional throughput while drastically cutting operational default rates and manual overhead.

• OPERATIONAL METRICS AT A GLANCE

\$1.4B+

PROCESSED VOLUME

Cumulative disbursements & collections

99.99%

PRODUCTION SLA UPTIME

Multi-region active-active cloud clusters

<100ms

P99 API LATENCY

Real-time mobile money settlement

5,000,000+

COMPLETED LOANS

Retail, MSME & agricultural credit

45+

DEPLOYMENTS

Commercial banks, SACCOs & MFIs

0.00%

BALANCE DRIFT

Cryptographic ledger validation

• CORE STRATEGIC PILLARS

1. Security-First Architecture

Every line of code complies with bank-grade standards: AES-256-GCM encryption at rest, TLS 1.3 in transit, hardware-backed multi-factor authentication, and immutable cryptographic audit trails.

2. Financial Inclusion Engine

Alternative credit scoring models ingest mobile wallet velocity, payment patterns, and utility records to safely underwrite thin-file, unbanked, and MSME borrowers traditionally rejected by legacy credit bureaus.

3. Mathematical Double-Entry

A deterministic double-entry accounting engine enforces mathematical balance consistency at the database engine level, eliminating discrepancies between core ledgers and third-party payment rails.

4. Institutional Partnership

We act as technology partners, providing 24/7/365 engineering telemetry, dedicated technical account managers, regulatory auto-reporting, and continuous infrastructure evolution.



● INDUSTRY PROBLEM & THE M&F SOLUTION

Modernizing Core Banking for the Digital Era

Traditional core banking systems were architected for branch-based cash deposits and nightly batch runs. Under high-frequency mobile lending demands, legacy infrastructure bottlenecks growth and inflates risk.

The Legacy Banking Bottleneck

- **Monolithic Batch Processing:** Balances reconcile overnight, causing dangerous multi-hour blind spots for mobile overdrafts and loans.
- **Manual Underwriting Backlog:** Human loan officers take 3 to 7 business days to review paper documentation and collateral records.
- **Frequent Balance Drift:** Discrepancies between external telco payment gateways and internal databases require costly manual adjustments.
- **Complex Regulatory Audits:** Compiling central bank prudential reports takes weeks of manual spreadsheet aggregation.

The M&F Modernization Paradigm

- **Event-Driven Microservices:** Real-time asynchronous transaction routing with sub-100ms disbursement execution to mobile wallets.
- **Automated Decisioning Engine:** Instant algorithmic credit evaluation utilizing traditional bureaus and alternative telecom telemetry.
- **Cryptographic Double-Entry Ledger:** Deterministic balance validation with zero balance drift and immutable audit logs.
- **Automated Regulatory Schemas:** 1-click generation of statutory Central Bank, IFRS 9 staging, and AML/CFT compliance filings.

● ARCHITECTURAL BENCHMARK COMPARISON

OPERATIONAL DIMENSION	LEGACY CORE BANKING ARCHITECTURE	M&F TECHNOLOGIES MIDDLEWARE
System Architecture	Monolithic relational database, single point of failure, scheduled nightly batch jobs.	Cloud-native containerized microservices, multi-region active-active clusters, 99.99% uptime.
Disbursement Speed	1 to 3 business days via manual branch approval and delayed clearing houses.	Sub-100ms instant disbursement directly to M-Pesa, Airtel Money, or direct RTGS bank rail.
Credit Underwriting	Manual risk committee review; limited strictly to historical formal credit bureau records.	Real-time machine decisioning tree; ingests mobile wallet velocity, utility data, and bureau records.
Ledger Consistency	Subject to balance drift during high-throughput network drops; manual ledger balancing.	Strict double-entry journal with cryptographic hash verification; 0.00% balance drift guarantee.
Integration & Extensibility	Proprietary closed protocols, slow enterprise bus adapters, months of custom code.	Enterprise JSON REST and GraphQL APIs, real-time webhooks, turnkey SDKs for web & mobile.
Statutory Compliance	Manual report generation; weeks of internal spreadsheet preparation before audits.	Automated regulatory auto-reporting; compliant schemas ready on demand for central bank auditors.

Zero Core Replacement Disruption: M&F Technologies can be deployed as an independent, standalone core banking system for digital lenders, OR integrated as an agile middleware layer on top of legacy cores (e.g., Finacle, T24, Orbit) without requiring risky multi-year migrations.



• MODULAR PRODUCT SUITE

The 10 Enterprise Technology Modules

Deploy the complete end-to-end lending stack or license individual microservices to enhance your existing banking architecture. Each module is independently scalable, highly available, and API-accessible.

MOD 01 Core Lending Engine & Ledger

Manages the entire loan lifecycle: application, underwriting, disbursement, servicing, amortized and reducing interest calculation, and final maturity. Built on mathematical double-entry accounting.

Double-Entry Zero Drift ACID Engine

MOD 02 Configurable Credit Scoring

Algorithmic decisioning engine aggregating traditional credit bureaus, alternative payment networks, and mobile wallet velocity. Risk teams can adjust score weights and cutoffs via a GUI.

Rule Matrix Alternative Data -18% Defaults

MOD 03 Collections & Recovery

Workflow-driven delinquency tracking. Segments overdue loans by risk tier, triggers automated SMS/email reminders, automated STK push payment prompts, and manages recovery officer queues.

Auto-STK Push Queue Routing +31% Recovery

MOD 04 Workflow & Approval Automation

Automates multi-tiered underwriting pipelines, instant AML/PEP sanctions screening, automated bank account verification, and committee approval matrices for high-value facilities.

AML/PEP Screening Policy Engine Instant Approvals

MOD 05 Developer Suite & API Gateway

Institutional REST and GraphQL APIs with sub-100ms P99 latency. Includes webhook subscription engine, multi-tenant rate limiting, comprehensive OpenAPI specifications, and sandbox testing.

<100ms Latency REST & GraphQL Webhooks

MOD 06 Regulatory Auto-Reporting

Automates compliance reporting for Central Banks and regulators. Generates standard IFRS 9 loan staging matrices, liquidity reports, capital adequacy returns, and statutory risk disclosures in 1 click.

IFRS 9 Models Central Bank Returns Audit-Ready

MOD 07 Multi-Currency & Cross-Border

Multi-currency ledger capable of holding, valuing, and disbursing across foreign and local currencies. Features live FX feed integrations, automated margin calculations, and regional treasury pooling.

Multi-Currency FX Rate Feeds Cross-Border

MOD 08 Document Management & OCR

Bank-grade encrypted cloud storage. Features automated OCR parsing for national identity cards, passport verification, bank statement extraction, e-signatures, and tamper-proof access logs.

OCR Extraction E-Signature Encrypted Vault

MOD 09 Real-Time Telemetry & Alerts

Comprehensive observability suite featuring Grafana and Prometheus telemetry. Real-time distributed tracing, endpoint health checks, server memory monitoring, and sub-minute incident alerts.

Grafana/Prometheus Tracing 24/7 Monitoring

MOD 10 Borrower & Field Agent Portals

Turnkey white-label borrower web applications and offline-first mobile apps for field credit officers. Captures biometric KYC and loan applications in remote areas with automated background sync.

Offline-First Biometric KYC White-Label



● INSTITUTIONAL ASSURANCE

Enterprise Security, Compliance & Governance

Lending infrastructure demands uncompromising trust. M&F Technologies implements institutional-grade security protocols, rigorous audit standards, and regulatory frameworks approved by leading financial authorities.

SOC 2 Type II Certified

Audited & Verified

Following an exhaustive 12-month independent audit conducted by a recognized third-party firm, M&F Technologies achieved SOC 2 Type II certification. The assessment verified our operational posture across all five Trust Services Criteria: **Security, Availability, Processing Integrity, Confidentiality, and Privacy.**

ENCRYPTION
AES-256-GCM

TRANSPORT
TLS 1.3 / PFS

ACCESS AUTH
Hardware MFA

TENANCY
Strict RBAC

● SECURITY & DISASTER RECOVERY MATRIX

Cryptographic Safeguards

- **Data at Rest:** Encrypted using AES-256-GCM with automated Hardware Security Module (HSM) key rotation.
- **Data in Transit:** Strict TLS 1.3 encryption with Perfect Forward Secrecy across all endpoints.
- **Immutable Audit Trails:** Every financial transaction is committed to an immutable append-only ledger journal.
- **Granular RBAC:** Least-privilege role-based access controls with session recording and mandatory MFA.

Resilience & Disaster Recovery

- **Multi-Region Active-Active:** Database nodes replicate in real time across isolated physical cloud zones.
- **RPO < 1 Minute:** Transaction journal replication ensures sub-minute recovery point objectives.
- **RTO < 5 Minutes:** Automated health probes trigger sub-five-minute cluster failovers during localized disruptions.
- **Canary Deployments:** Production software releases follow automated canary testing with zero system downtime.

● REGULATORY ALIGNMENT & DATA PRIVACY

Central Bank Compliance

Fully compliant with Central Bank prudential guidelines, capital adequacy metrics, and interest rate transparency directives.

Data Protection (ODPC & GDPR)

Strict adherence to the Kenya Data Protection Act 2019 and European GDPR. Provides data sovereignty, localized hosting, and consent management.

AML / CFT Oversight

Automated screening against global OFAC, UN, and PEP sanction watchlists with automated transaction monitoring and suspicious activity flags.



• PROVEN TRACK RECORD

Institutional Deployments & Client Impact

M&F Technologies powers leading commercial banks, regional cooperative SACCOs, and digital micro-lenders. Our deployments deliver quantifiable gains in operational speed, risk mitigation, and balance sheet performance.

Apex International Bank

1.2M ACCOUNTS

Commercial & Retail Banking • Nairobi, Kenya

Challenge: Outdated legacy core banking caused 4-day loan turnaround times, leading to high drop-off rates for digital retail facilities.

Solution: Deployed M&F Core Lending Engine and API Middleware to automate origination and direct-to-mobile disbursements.

QUANTIFIABLE IMPACT:

- 74% reduction in loan approval turnaround (from 4 days to 4 minutes)
- 18% lower non-performing loans (NPL) via alternative credit scoring

Pioneer Teachers SACCO

65K MEMBERS

Credit Union / Cooperative • Nairobi, Kenya

Challenge: Manual spreadsheet tracking for overdue member facilities resulted in mounting delinquency and slow collection cycles.

Solution: Integrated M&F Collections & Delinquency Management with automated SMS schedules and mobile money STK push prompts.

QUANTIFIABLE IMPACT:

- 31% recovery rate improvement in the first 90 days
- 100% automated payment reconciliation with zero manual spreadsheets

Sunrise Micro-Finance Group

OFFLINE-FIRST

Rural & Agricultural MFI • Mombasa, Kenya

Challenge: Field credit officers operating in remote agricultural zones lacked reliable internet, slowing KYC capture and disbursements.

Solution: Equipped 200+ field officers with the M&F Offline-First Mobile App with encrypted SQLite storage and biometric capture.

QUANTIFIABLE IMPACT:

- 60% boost in daily field officer application throughput
- Automated sync on network reconnect with zero corrupted records

Equatorial United Bank

HIGH VELOCITY

Commercial Bank • Kampala, Uganda

Challenge: Bank experienced server crashes during month-end salary loan peaks, causing angry borrower backlogs and SLA penalties.

Solution: Deployed M&F Configurable Decisioning Engine with dynamic query queuing and active-active load balancing.

QUANTIFIABLE IMPACT:

- Processed 45,000+ credit scoring queries/hour during peak windows
- 100% uptime with zero server lockups across 24 consecutive months

• INSTITUTIONAL CLIENT SEGMENTS SERVED

Commercial & Retail Banks

Tier-1 & Tier-2 institutions modernizing retail lending portfolios

Credit Unions & SACCOs

Community cooperatives digitizing member loan applications

Digital MFIs & Fintechs

High-velocity digital credit platforms requiring sub-second APIs



• CORPORATE GOVERNANCE & DELIVERY

Leadership & Implementation Methodology

Our team brings deep expertise in core banking systems, distributed ledger architecture, and enterprise compliance. We back every engagement with a structured, low-risk 4-phase deployment methodology.

• EXECUTIVE LEADERSHIP & ENGINEERING CULTURE

Musa Mutindi

FOUNDER & CHIEF EXECUTIVE OFFICER

Leads M&F Technologies with a focus on mission-driven financial infrastructure expanding credit access. Oversees institutional strategy, banking partnerships, and product direction.

Distributed Engineering Group SYSTEMS & LEDGER ARCHITECTS

Comprises senior database reliability engineers, distributed systems developers, and security experts operating across five time zones to ensure sub-100ms response SLAs and 99.99% uptime.

Compliance & Risk Advisory REGULATORY FRAMEWORK COUNCIL

Specialized banking compliance officers and regulatory legal experts who shape our Central Bank reporting schemas, IFRS 9 impairment models, and SOC 2 Type II audit readiness.

• THE 4-PHASE INSTITUTIONAL IMPLEMENTATION ROADMAP

PHASE & TIMELINE	KEY OBJECTIVES	CORE TECHNICAL ACTIVITIES	INSTITUTIONAL DELIVERABLE
PHASE 1 Weeks 1 – 2	Architecture Review & Gap Analysis	Audit existing core banking interfaces, map data dictionaries, define risk weight trees, and configure loan products.	Architecture Blueprint & Data Mapping Specification
PHASE 2 Weeks 3 – 4	Sandbox Provisioning & API Integration	Deploy dedicated multi-tenant sandbox environment; configure payment gateway webhooks (M-Pesa/Airtel/RTGS); connect credit bureaus.	Functional Integration Sandbox & Test Suite
PHASE 3 Weeks 5 – 6	Shadow Ledger & Parallel Validation	Run M&F double-entry ledger in parallel with legacy system; verify 0.00% balance drift; perform user acceptance testing (UAT).	UAT Sign-off & Ledger Audit Certification
PHASE 4 Week 7 Onward	Production Cutover & 24/7 SLA	Live production traffic cutover; staff enablement workshops; continuous monitoring with dedicated 24/7 Tier-1 engineering support.	Live Production System & Guaranteed 99.99% SLA

Institutional Support & SLA Commitment

Guaranteed 15-minute response for critical P1 incidents • 24/7/365 Tier-1 engineering operations • Dedicated Technical Account Manager

99.99% SLA



• INSTITUTIONAL INQUIRIES & PROCUREMENT

Corporate Directory & Engagement

Whether you are a commercial bank seeking to modernize your lending stack, a credit union expanding member access, or a fintech platform integrating institutional APIs, M&F Technologies is ready to partner.

Corporate Headquarters

M&F Technologies Limited

The Pavilion, 4th Floor
Lower Kabete Road, Westlands
Nairobi, Kenya

Operating Hours: Monday – Friday: 08:00 – 17:00 EAT

Global Network Operations: 24/7/365 Continuous Monitoring

Institutional Communication

General & Corporate Inquiries:
info@mfttechnologies.org

Institutional Hotlines & Support:
Main Support: +254 748 329 410
Client Hotline: +254 701 547 500

Official Web & API Resources:
<https://mfttechnologies.org> • API Docs: /docs

• INSTITUTIONAL RFP & PROCUREMENT PROTOCOL

Request for Proposals (RFP) & Proof of Concept (PoC)

Financial institutions conducting formal procurement, core modernization feasibility studies, or security vendor assessments are invited to submit RFPs or schedule an executive technical briefing.

1. Architecture Briefing

Schedule a 45-minute technical deep dive with our systems engineering leads.

2. Custom Sandbox

Receive a private staging sandbox tenant pre-configured with your lending products.

3. Compliance Pack

Request our SOC 2 Type II audit report, pen-test summaries, and SLA agreements.

M&F TECHNOLOGIES LIMITED

Certified Institutional Technology Provider • SOC 2 Type II Audited • Nairobi, Kenya

OFFICIAL DOSSIER

REF: MF-CORP-2026-V1

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